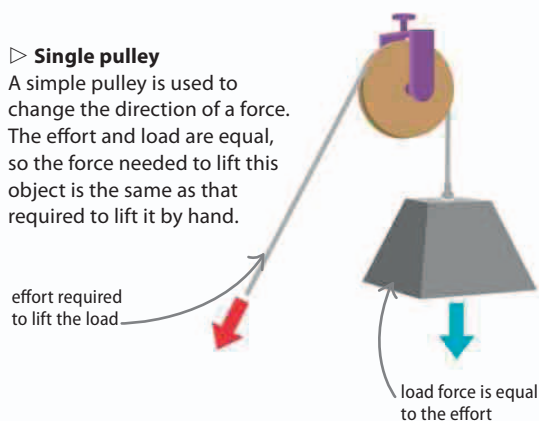


Pulleys

Compound pulleys are good examples of the way machines create a mechanical advantage—amplifying a small effort into a large force capable of lifting big loads. The double pulley is the simplest type. Single pulleys do not create a mechanical advantage, but they allow a force to be applied in a different direction.

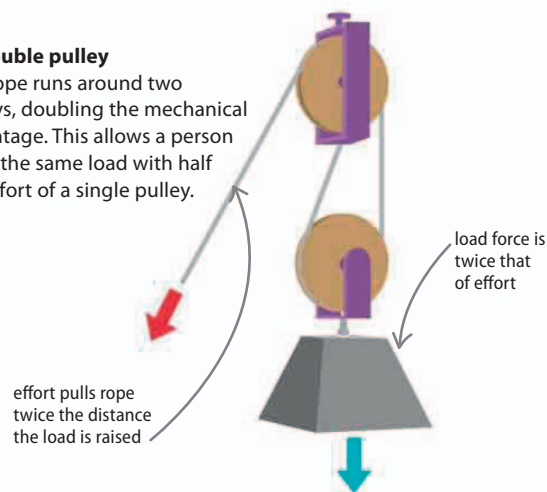
▷ Single pulley

A simple pulley is used to change the direction of a force. The effort and load are equal, so the force needed to lift this object is the same as that required to lift it by hand.



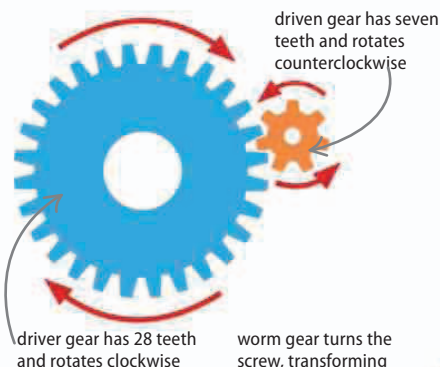
▷ Double pulley

The rope runs around two pulleys, doubling the mechanical advantage. This allows a person to lift the same load with half the effort of a single pulley.



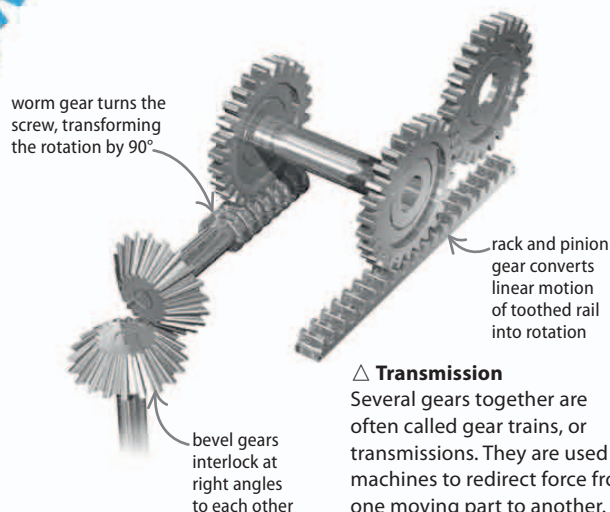
Gears

When wheels are given interlocking teeth they become cogs or gears, which are used to transmit a turning force, or torque. The magnitude of the transmission depends on the gear ratio, a comparison of the number of teeth on each gear. For example, when the driver gear (moved by the effort force) has twice the number of teeth of the driven gear, this second gear rotates twice as fast and with half the torque.



◁ Gear ratio

To calculate the gear ratio of the gears below, divide the number of teeth of the driver (left; 28 teeth) by that of the driven gear (right; seven teeth). The answer is four, which means that the smaller, driven gear will turn four times faster than the speed of the larger, driver gear.



△ Transmission

Several gears together are often called gear trains, or transmissions. They are used in machines to redirect force from one moving part to another.

REAL WORLD

Excavator

Construction machines, such as this excavator, show how simple machines are combined. The digger moves on tracks, which are driven by wheels acting as pulleys at each end. The shovel uses a wedge to cut into the ground, and moves using hydraulic levers.

